

Assignment - Potential Student Predictions

Introduction

The five bootcamps offered by the university Super Star University are training programs that help students equip themselves with skills such as utilising programming languages, spreadsheet editors, and media marketing. Even though there is an 18 per cent enrollment in these programs, it was discovered that it is labour-intensive and time-consuming to contact random students. Therefore, for the new bootcamp program for data science, enrollments can be improved by identifying segments of students that are likely to accept the offer. This is done by building predictive models with machine learning techniques with existing and future data collected by the university to identify and predict students who should be contacted to increase the chances of enrollment in the programs.

Three models that will be used for the objective are the k-nearest neighbours algorithm (kNN), decision tree, and artificial neural network (ANN). The kNN algorithm is used for classification tasks. It identifies the closest data points to a given query point and makes predictions based on the majority class or average outcome of these neighbours. Decision trees are also used for classification by splitting the data into subsets based on the conditions of the input features and creating a tree-like model of decisions. ANN processes data in a way that is inspired by the human brain and consists of neurons in input, hidden, and output layers to conduct classification tasks. The performance of these will be assessed using evaluation metrics to determine which model is the best.

The dataset contains variables including the alumni's basic personal information, activity details with the salesperson, occupation, student's results tier, and their monthly income. The target variable for this dataset is taken which has binary outcomes that indicate whether the student has attended a bootcamp. If the value shows one, it indicates that the student has attended at least one of the bootcamps. If the value shows zero, it indicates that the student did not attend any of the bootcamps.

Variable	Description
StudentID	Unique student ID of the alumni
Taken	Whether the student has attended a bootcamp or not (0: No, 1: Yes)
Age	Age of alumni.
TypeofContact	How alumni was contacted (University Called or Self Inquiry)
CGPATier	CGPA Tier is the student result. The categories are ordered i.e. Tier 1 > Tier 2 > Tier 3
DurationOfPitch	Duration of the pitch by a salesperson to the alumni
Occupation	Occupation of alumni
Gender	Gender of alumni
NumberOfFollow ups	Total number of follow-ups has been done by sales person after sales pitch
BootcampPitched	Bootcamp pitched by the salesperson
MaritalStatus	Marital status of alumni
PitchSatisfaction Score	Sales pitch satisfaction score
OwnCar	Whether the alumni own a car or not (0: No, 1: Yes)
Designation	Designation of the alumni in the current organization
MonthlyIncome	Gross monthly income of the alumni

Table 1 - Description of variables

Exploratory Data Analysis

Variable	Count	Mean	Std	Min	Max
StudentID	4888.0	202443.500	1411.188	200000.0	204887.0
Taken	4888.0	0.188	0.391	0.0	1.0
Age	4662.0	37.622	9.316	18.0	61.0
CGPATier	4888.0	1.654	0.917	1.0	3.0
DurationOf Pitch	4637.0	15.491	8.520	5.0	127.0
NumberOf Followups	4843.0	3.708	1.003	1.0	6.0
PitchSatisfa ctionScore	4888.0	3.078	1.366	1.0	5.0
OwnCar	4888.0	0.620	0.485	0.0	1.0

Table 2 - Summary Statistics

The summary statistics provide the mean, standard deviation, minimum, and maximum values in the data for each interval variable. The maximum value for duration of pitch is 127, which indicates an anomaly or an outlier with an extreme value that deviates far from the expected duration.

Graphical displays such as histograms and boxplots can be used to visualise the behaviours of the data such as their distribution, variability, central tendency, and outliers of the data. Histograms display the distribution of the variables to show if the data is normally distributed or has skewness in the distribution. Boxplots are able to clearly show outliers in variables. The box in the boxplot shows the interquartile range, which is the range between the first quartile and the third quartile. The whiskers are the lines at the top and bottom of the box that represent the range of the data. Points that are extremely far away from the whiskers are considered outliers.

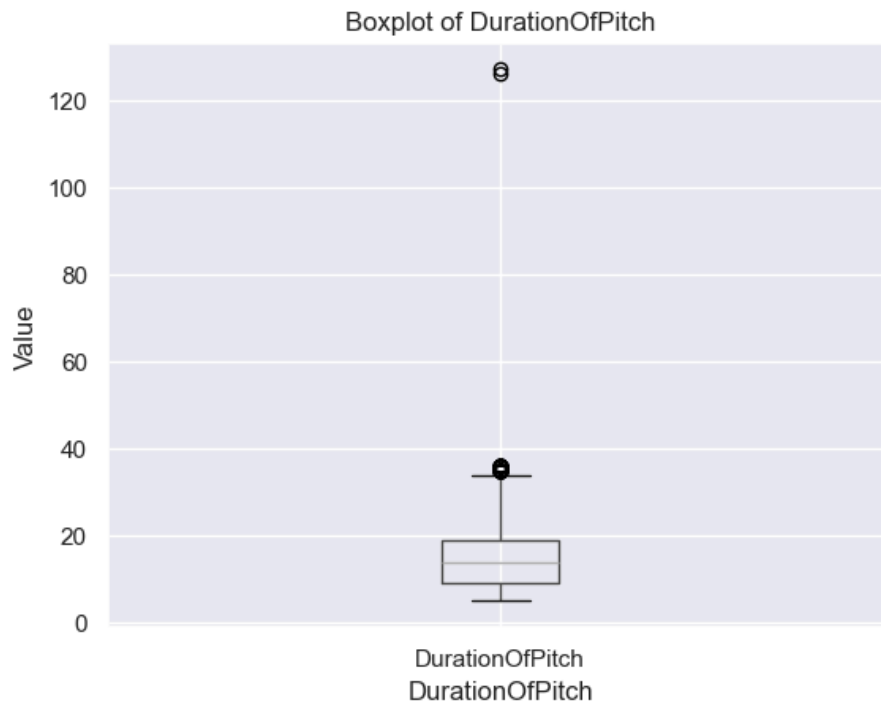


Figure 1 - Boxplot of duration of pitch

Based on the boxplot for the duration of pitch variable, there are some outliers above the value of 120, which is unusually high compared to the rest of the data. Therefore, these outliers may need to be removed. This is done by specifying a cutoff range of values above 120 to remove the outliers during data preprocessing



Figure 2 - Boxplot of monthly income

Based on the boxplot for the monthly income variable, some outliers deviate far away from the whiskers on the top and the bottom. A cutoff range of values below 10000 is set to remove the outliers at the bottom, and a cutoff range of values above 80000 is set to remove the outliers at the top during data preprocessing.

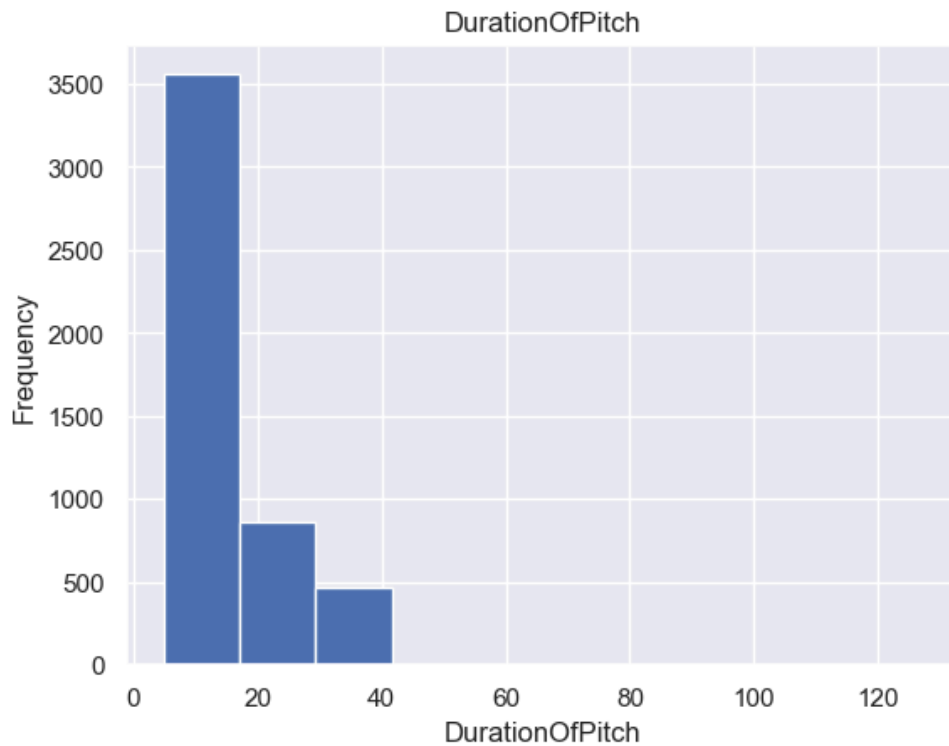


Figure 3 - Histogram of duration of pitch

Based on the histogram for the duration of pitch variable, the distribution is positively skewed to the right. This might be due to the outliers with extreme values majorly contributing to the skewness in the distribution.

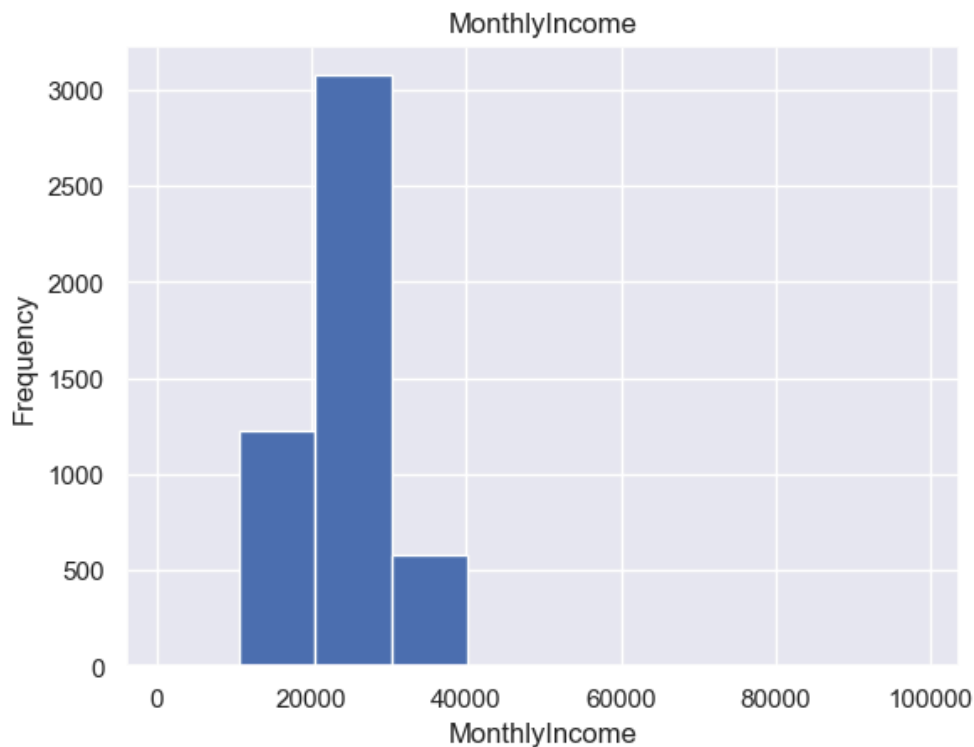


Figure 4 - Histogram of monthly income

Based on the histogram for the monthly income variable, the distribution is positively skewed to the right. This also might be caused by the outliers with extreme values majorly contributing to the skewness in the distribution.

Data Preprocessing

Before conducting analysis and building models, the data has to be preprocessed. A dataset that contains poor-quality data may negatively impact the performance and accuracy of predictive models. Therefore, data preprocessing is a crucial step before developing models. After checking the unique data of each categorical variable, it is found that the variable gender has three unique data including “Fe Male”. This may be due to an input or data collection error and it should be replaced with Female. This ensures that gender has only two unique inputs.

Interval variables such as age, duration of pitch, number of follow ups, and monthly income contain missing values. Since there is a considerably large percentage of data that has missing values out of 4888 observations, their values can be replaced by using imputation to preserve information in the data. Having a significant portion of missing values may lead to incorrect predictions and misleading model accuracy, so handling those missing values allows the analysis to be more reliable and accurate. The missing values in the interval variables age, duration of pitch, number of follow ups, and monthly income will be replaced with the mean value using imputation. Similarly, the missing values in the categorical variable type of contact are replaced with the mode or the category level with the most frequency.

The outliers that were spotted have been removed using cutoff ranges specified during exploratory data analysis. After the outliers in the duration of pitch and monthly income variable have been removed, the skewness in their distributions reduced significantly.

The data types of some columns such as age, number of followups, duration of pitch, and monthly income variable are string type. Their data types have been converted to integer type to ensure that functions and models that require numerical inputs can proceed without any issues.

When variables are on different scales, algorithms that rely on distance metrics such as kNN can be biased towards variables with larger scales. To ensure that all variables have the same scale so that each feature contributes equally to the result, normalisation can be conducted to transform the data to fit in the range of zero to one. Proper feature scaling helps to stabilise predictions across different k values. The formula of the normalisation involves subtracting the original value by the minimum value of the data in the variable, then divide by the denominator of the maximum value subtracted by the minimum value.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Machine learning models require the categorical data to be represented as numerical inputs so that algorithms can be utilised validly. Therefore, categorical variables have to be encoded to convert them into numerical inputs using label encoding. Label encoding assigns unique integer values for each category level in categorical variables. The code shown in the Python script utilises the `loop` function that iteratively encodes each categorical variable to transform them into numerical inputs.

The correlation heat map shows the degree of correlations between variables. The correlation between variables is considerably small. If there are correlations between variables, the matrix becomes singular, therefore there is no inverse matrix and vectors may not be independent. When this is the case, multicollinearity can occur and may result in unreliable performance and misleading interpretations. Since there are no significant correlations between the variables, removing variables from analysis and model building is unnecessary and all of the variables will be used in the models.

Model training and validation

kNN

Before building the machine learning models, an 80 to 20 data partitioning ratio is applied, where 80% of the data is used for train and 20% for test. Having 80% of the data for train set ensures that there is enough data for the model to learn and 20% for test set ensures that there is an adequate assessment of the model's performance to new data. The random state is assigned with a value as a set seed so that the results are reproducible. Shuffling the data before splitting leads to a random distribution of data for train and test sets to be representative of the overall dataset and avoid potential biases.

The first machine learning model developed is kNN. kNN algorithms require a specified value of k . The value k represents the number of closest data points required to be included in the majority voting that determines the classification of new data points. Euclidean distance is used to identify the closest data points. The grid search function allows testing of different hyperparameter values from 1 to 30 using `np.arange()`, which also includes identifying the best parameter settings. The best parameter is found to be $k = 1$, which means that the best kNN model uses one nearest neighbour for making predictions. The accuracy of the model is shown to be 100%, which shows that it correctly predicts the classes of the observations every single time.

kNN algorithm with one nearest neighbour for predictions assigns the class of the one nearest data point to the new observation. The 1-nearest neighbour classifier may result in high variance and low bias, which is indicated by a perfect accuracy score. The bias is low since with 1-nearest point, the model may fit too closely to the training data. The variance is high since optimizing on only 1-nearest point means that there is a high chance of containing noise in the data. Therefore, 1-nearest point may not be an ideal kNN model.

A confusion matrix is a table used to evaluate the performance of the classification model. It displays comparisons between the actual target values with the predicted values by showing the true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). True positives are the correctly predicted positive cases or cases of enrollments. False positives are the incorrectly predicted positive cases or cases without enrollments. True negatives are the correctly predicted negative cases. False negatives are incorrectly predicted negative cases. These values help inspect the performance of the model using different metrics such as accuracy, sensitivity, specificity, and precision. The accuracy indicates the proportion of correctly predicted cases to the total number of cases, or both the true positives and true negatives out of all cases. Sensitivity defines the proportion of correctly predicted positive cases to all actual positive cases or the true positives out of all positive instances which also include false negatives. Specificity shows the correctly predicted negative cases to all actual negative cases or the true negatives out of all negative instances which also include false positives. Precision specifies the proportion of correctly predicted positive cases among all cases predicted as positive, or the true positives out of all predicted positives including the false positives.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Precision = \frac{TP}{TP + FP}$$

		Actual Cases	
		Positive (1)	Negative (0)
Predicted Cases	Positive (1)	180 (TP)	0 (FP)
	Negative (0)	0 (FN)	797 (TN)

Table 3 - Confusion Matrix for kNN model with 1-nearest neighbour

The confusion matrix table shows that there are no incorrectly classified observations and all observations have been predicted correctly. This suggests that it has a perfect accuracy score. A model with such accuracy may indicate overfitting and may not perform well on future unseen data. The sensitivity and specificity are also 100%, which indicates that the model is equally perfect at identifying positive and negative cases. This may seem ideal, but this is unrealistic in terms of practicality and can cause overfitting, which may lead to poor model performance on future unseen data. Precision is found to also be 100%, which represents absolute reliability in predicting positives, and it is again hardly feasible in practical applications.

Decision Tree

The second model developed is the decision tree. It is a tree-like model that splits the dataset into subsets based on the splitting rules determined by the value of input features. The root node consists of the entire dataset, and then it splits until the tree reaches its maximum depth and displays the leaf nodes representing the final decision results. The grid search function allows testing of different maximum depths from 1 to 30. The best parameter is found to be a maximum depth of 16. The accuracy of the model is shown to be 87.7%, which is considerably high.

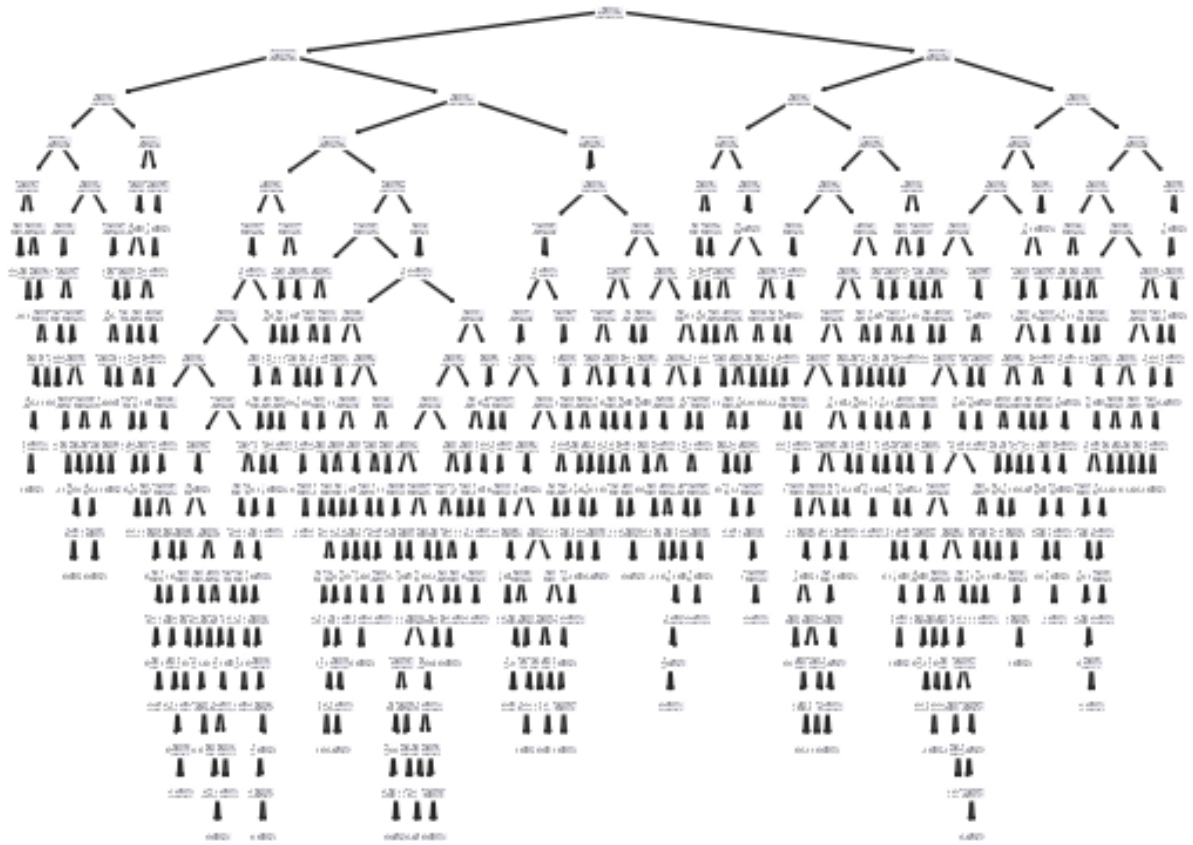


Figure - Decision tree diagram

		Actual Cases	
		Positive (1)	Negative (0)
Predicted Cases	Positive (1)	113 (TP)	53 (FP)
	Negative (0)	67 (FN)	744 (TN)

Table 4 - Confusion Matrix for decision tree model with a maximum depth of 16

The confusion matrix table shows that a majority of positive and negative cases have been predicted correctly. This suggests that it has a relatively high accuracy. The sensitivity is 62.8% and the specificity is 93.4%. This indicates that the model has a moderately good performance at identifying positive cases, but is way better at identifying negative cases. Precision is found to be 68.1%, which shows moderate reliability in predicting positive cases.

ANN

The last machine learning model developed is ANN. The structure of ANN is composed of input, hidden, and output layers. The neurons in input layers receive input data and represent themselves as features of the data. The neurons in hidden layers process the inputs and determine the importance of information provided by the features in the data and apply activation functions. Activation functions such as sigmoid, hyperbolic tangent, and rectified linear units introduce non-linearity into the model so that it can learn complex relationships within the data. The results are passed to the output layer and then it provides the final prediction results.

The grid search function allows testing of different numbers of neurons in hidden layers from 1 to 30. The best parameter is found to have 12 neurons in the hidden layer. The accuracy of the model is shown to be 81.7%, which is high but relatively lower than the decision tree.

		Actual Cases	
		Positive (1)	Negative (0)
Predicted Cases	Positive (1)	2 (TP)	1 (FP)
	Negative (0)	178 (FN)	796 (TN)

Table 5 - Confusion Matrix for artificial neural network model

The majority of observations are predicted as negative cases. There is almost an absence of observations that are predicted as positive cases. This suggests that the model has a high specificity but a very low sensitivity. This indicates that the model perfectly identifies negative cases, but is almost futile in identifying positive cases. The dataset split for train and test may be imbalanced for this neural network model and may cause the neural network to be conservative in determining positive cases. Precision is found to be 66.7%. However, since there are very little cases that are predicted as positive, the precision metric may not be a good indicator of the model's performance.

Model Selection

Based on the three machine learning models developed, the decision tree has the highest accuracy and is the favoured model to be used to predict the prospective students that are likely to enroll in the data science program. The kNN model that has been built has a parameter of 1-nearest neighbour and may be prone to overfitting due to its perfect accuracy. The ANN model has a structure that is highly complex and interpretations of output produced by this model can be difficult for the management to utilise. The decision tree diagram can provide visual representations of the splitting rules based on the conditions of variables. Therefore, these splitting rules are indicative of the important variables that influence student enrollment decisions. The model also provides clear decision paths that lead to successful enrollments.

Conclusion

Decision Trees provide significant advantages in understanding and predicting prospective student enrollments for the new bootcamp program at 'Super Star University.' The visualisations and insights offered by the decision tree model allow us to pinpoint specific factors that influence enrollment decisions, which can help the management determine student segments who are likely to accept offers. The management can prioritise targeting students based on these factors to increase the chances of student recruitment with minimal labour and time spent contacting students. Leveraging insights provided by the model helps to personalise outreach and communication with prospective students to deliver appealing and high-quality pitches.

The management may also provide comprehensive training for the salesperson in pitching to reduce the amount of time spent and improve student engagement. During their approach to contacting students, they should highlight program features and benefits that align with their interests to significantly increase engagement and improve enrollment rates. Understanding the purpose that alumni can achieve by enrolling into the programs helps to improve personalisation in communications and services. The findings also provide indications for improving targeted marketing campaigns to attract more prospective students. These suggestions can contribute to higher student enrollment rates in the program and help to support the institution's growth and success.